



NORTH SYDNEY BOYS HIGH SCHOOL

2023 YEAR 12 HSC ASSESSMENT TASK 3

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – **3 hours**
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Ms Sarofim
- ☐ Ms Fu
- ☐ Mr Lin
- ☐ Mr Ireland
- ☐ Dr Vranesevic

Student Number:

(To be used by the exam markers only.)

Question No	1-10	11- 30	Total
Mark	<u>10</u>	<u>90</u>	<u>100</u>

Outcomes – Section I

[illegible]

Section I

10 marks

Allow about 15 minutes for this section. Use the multiple-choice answer sheet

1 The period of the function $f(x) = 2 \tan(4x - \frac{\pi}{3})$ is

- A $\frac{\pi}{2}$
- B $\frac{\pi}{3}$
- C $\frac{\pi}{4}$
- D π

2 For what values of x is the curve $f(x) = x^4 - 2x^3$ concave down?

- A $[0, \frac{3}{2}]$
- B $(0, \frac{3}{2})$
- C $(0, 1)$
- D $[0, 1]$

3 Which expression is the derivative of $\cos^2 3x$ when differentiated with respect to x ?

- A $-6 \sin 3x \cos 3x$
- B $-2 \sin 3x \cos 3x$
- C $2 \sin 3x \cos 3x$
- D $6 \sin 3x \cos 3x$

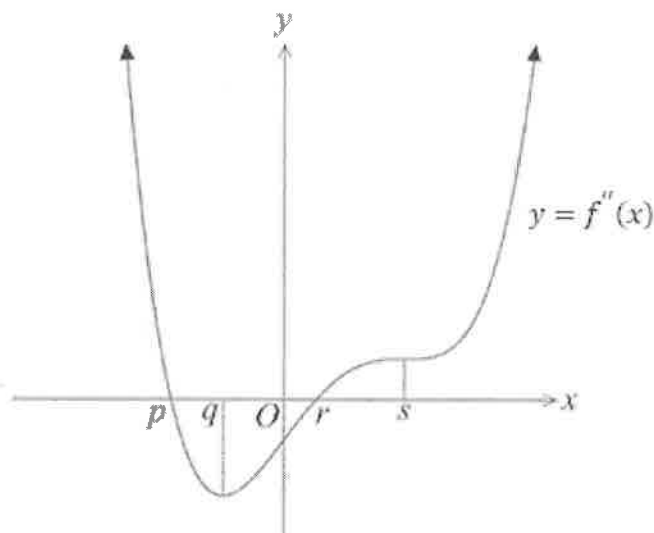
4 The amount M of a drug present in the blood after t hours is given by

$$M = 9t^2 - t^3 \text{ for } 0 \leq t \leq 9.$$

When is the amount of drug in the blood increasing most rapidly?

- A $t = 0$
- B $t = 9$
- C $t = 6$
- D $t = 3$

- 5 The diagram shows the graph of $y = f''(x)$ for the function $f(x)$.



For what value of x does the function $f'(x)$ have a maximum turning point?

- A $x = p$
- B $x = q$
- C $x = r$
- D $x = s$

- 6 The table below shows the probability distribution of a discrete random variable X which has mean -0.45 .

x	-2	-1	0	1	2
$P(X = x)$	0.1	a	0.2	0.15	b

What are the values of a and b ?

- A $a = 0.2, b = 0.35$
- B $a = 0.3, b = 0.25$
- C $a = 0.4, b = 0.15$
- D $a = 0.5, b = 0.05$

- 7 The graph of the function $y = f(x)$ is known to have a minimum turning point at $P(6, -4)$. Therefore the graph of $y = -f(-2x)$ will have a maximum turning point at

- A $(3, 4)$
- B $(-3, 4)$
- C $(-3, -4)$
- D $(-12, 4)$

- 8 Let X and Y be two events such that $P(X) = 0.5$, $P(Y) = 0.6$, and $P(Y|X) = 0.7$.

Which of the following statements is FALSE?

- A $P(X|Y) < P(Y|X)$
- B X and Y are independent events
- C $P(X \cap Y) = 0.35$
- D $P(X \cup Y) = 0.75$

- 9 What are the domain and range of the function $f(x) = \ln(x+1) - \sqrt{4-x^2}$?

- A domain $(-1, \infty)$ range $(-\infty, \infty)$
- B domain $(-1, \infty)$ range $(-\infty, \ln 3]$
- C domain $(-1, 2]$ range $(-\infty, \infty)$
- D domain $(-1, 2]$ range $(-\infty, \ln 3]$

- 10 Which of the following is the correct function value at the minimum turning point of

$$f(x) = (x - 2021)(x - 2022)(x - 2023)$$

- A $\frac{1}{\sqrt{3}}$
- B $-\frac{1}{\sqrt{3}}$
- C $-\frac{2\sqrt{3}}{9}$
- D $\frac{2\sqrt{3}}{9}$

2023 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Class and Teacher:

Student Number:

Mathematics Advanced Section II

90 marks

Attempt Questions 11-30

Allow about 2 hours and 45 minutes for this section

Instructions

- Write your Student Number at the top of this page.
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	2023 Y12 Adv - Assessment Task 3 – Outcomes – Section II											
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11							/3					
12		/2										
13			/3									
14									/3			
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16					/3							
17							/4					
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24				/5								
25								/6				
26				/4								
27				/2								
28					/4							
29									/5			
30						/7						
Total	/5	/2	/3	/16	/7	/15	/7	/6	/8	/21	/90	

Question 11 (3 marks)

The third term of an arithmetic series is 32 and the sixth term is 17.

- (i) Find the common difference 1

- (ii) Find the sum of the first ten terms. 2

Question 12 (2 marks)

If $\sin \theta = \frac{3}{5}$ and $\tan \theta < 0$, find the exact value of $\cos \theta$ 2

Question 13 (3 marks)

Given that $2\log_e(x^2y) = 3 + \log_e x - \log_e y$, express y in terms of x in simplest terms. **3**

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Question 14 (3 marks)

Find the equation of the normal to $y = e^{\cos x}$ at the point where $x = \frac{\pi}{2}$ **3**

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Question 15 (7 marks)

(i) Find $\int \sec^2 3x \, dx$

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(ii) Find $\int \frac{1}{(3x+2)^4} dx$

2

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(iii) Evaluate $\int_0^1 e^{-2x} dx$ exactly.

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(iv) Find $\int \frac{x^2}{x^3-5} dx$

2

Question 16 (3 marks)

The curve $y = \sin x$ is stretched horizontally by a factor of 2, then it is shifted $\frac{\pi}{2}$ units right, then it is stretched vertically by a factor of 3 and reflected in the x -axis.

What equation describes the final curve after this sequence of transformations?

3

Question 17 (4 marks)

A new brand of electric bicycle is introduced to the market and 18,000 are sold in the first month. Each month thereafter, the sales are 70% of the sales in the previous month.

- (i) In which month will monthly sales first drop below 1000 per month? **2**

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- (ii) How many bicycles are sold in total in the first year? **1**

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- (iii) How many bicycles are eventually sold altogether? **1**

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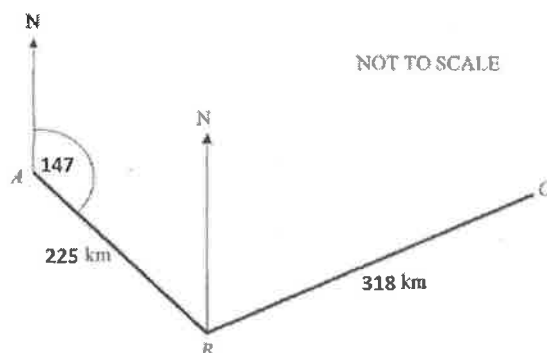
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Question 18 (5 marks)

A ship sails 225 km from Adhiban Island on a bearing of 147 degrees and arrive at Port Bologan to pick up some cattle. It then progresses to its destination, Port Cramling, a distance of 318 km on a bearing of 071 degrees.



- (i) Show that $\angle ABC = 104^\circ$
(you may write on the diagram above)

1

- (ii) Show that the distance AC is approximately 431.7 km.

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- (iii) The return trip is a straight line back to Adhiban Island and not passing through Port Bologan. Find the bearing that the ship must take to go straight from Port Cramling to Adhiban Island.

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Question 19 (5 marks)

Mischa likes to drink pearl milk tea at work. The number X of teas she drinks each day is a random variable with probability distribution given by:

x	0	1	2	3
$P(X = x)$	0.1	0.2	0.3	0.4

- (i) What is the expected value $E(X)$?

1

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- (ii) What are the variance, $\text{Var}(X)$, and the standard deviation, σ ?

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- (iii) Mischa is at work on two successive days. What is the probability that she drinks the same number of pearl milk teas on both days?

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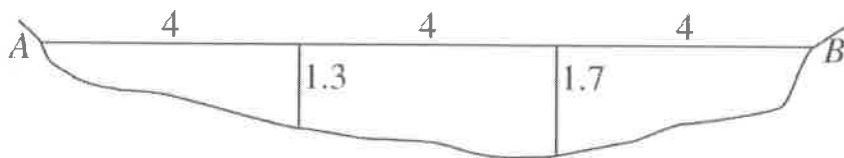
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Question 20 (4 marks)

The diagram below shows the cross-section of a stream with the depths of the stream shown in metres at 4 metre intervals. The creek is 12 metres wide.



- (i) Use the trapezoidal rule to approximate the area of the cross-section.

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- (ii) If water flows through this part of the stream at a speed of 0.5 metres/sec, calculate the approximate volume of water that flows past this section in 1 hour.

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Question 21 (8 marks)

Consider the function $y = x^3 - 9x^2 + 24x$.

- (i) Find all stationary points and determine their nature.

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(ii) Find the point of inflection.

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(iii) Sketch the curve showing all important features.

2

Question 22 (5 marks)

The line L is the tangent to the curve $y = x^3 + 7$ at $x = 2$.

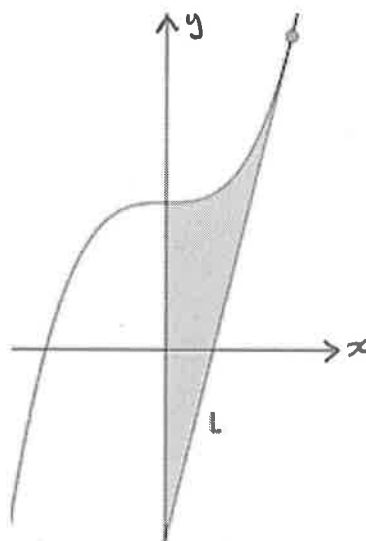
- (i) Show that the equation of the tangent L is $y = 12x - 9$

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- (ii) Find the area bounded by the y-axis, the tangent L, and the curve $y = x^3 + 7$

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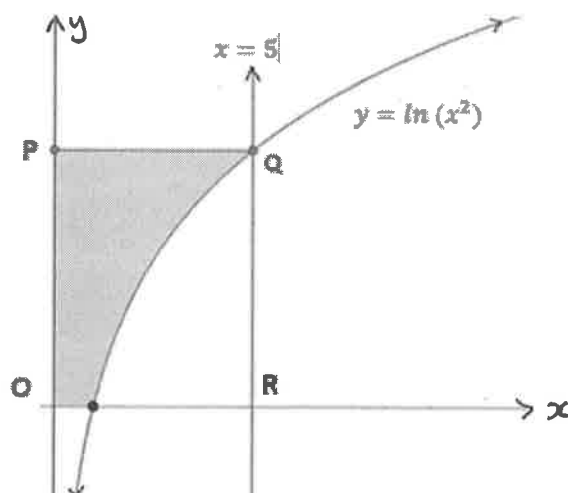
Question 23 (5 marks)

- (i) Show that $\frac{d}{dx} (x \ln x - x) = \ln x$ 1

- (ii) Hence or otherwise find $\int \ln x^2 dx$ 1

- (iii) The graph shows the curve $y = \ln x^2$, ($x > 0$) which meets the line $x = 5$ at Q. 3

Using your answers from parts (i) and (ii), or otherwise,
find the area of the shaded region.



Question 24 (5 marks)

A six-sided die is biased so that the number 5 occurs twice as often as any other number.

- (i) The die is rolled once. Show that the probability that an odd number occurs $\frac{4}{7}$. **1**

- (ii) If the biased die is rolled twice, find the probability that the sum of the uppermost numbers is seven. **2**

The biased die is now rolled together with TWO fair six-sided dice.

(iii) What is the chance that at least two odd numbers are uppermost?

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Question 25 (6 marks)

The outside temperature (in degrees Celsius) on a certain day was modelled

by $T = 12 + 7\sin\left(\frac{\pi t}{12}\right)$ where t is the number of hours after 6am.

- (i) What is the maximum temperature in the day? **1**

- (ii) Sketch a graph of the function $T = 12 + 7\sin\left(\frac{\pi t}{12}\right)$ for $0 \leq t \leq 24$ **2**

- (iii) Between what times during the day is the temperature 15°C or above? **3**

Question 26 (4 marks)

Mrs McCrone walks her three labradoodles at Balmoral Beach every Saturday morning. The dogs are poorly behaved: if a stranger pats them, the chance that the white dog bites him is $\frac{1}{20}$, the chance that the brown one bites him is $\frac{1}{10}$, and the chance that the deranged black one bites him is $\frac{1}{2}$.

- (i) If a stranger selects one of the dogs at random and pats it, what is the chance they will be bitten? **2**

- (ii) Given that a stranger pats one of the dogs and is bitten, what is the probability that it was the black one they patted? **2**

Question 27 (2 marks)

Edward plays a game in which he has a probability p of winning, probability q of losing, and probability r of moving to the next round ($p + q + r = 1$).

What is his probability of eventually winning, in terms of p and q ?

2

[illegible]

Question 28 (4 marks)

The point $A(6, 1)$ lies on $h(x)$. The tangent at A is $y = \frac{x}{6}$. Point B is the image of A on the function $g(x) = 3h(2x + 4)$.

(i) Show that B has coordinates $(1, 3)$.

1

[illegible]

(ii) Hence find the equation of the tangent to $g(x)$ at B .

3

[illegible]

Question 29 (5 marks)

Consider the function $f(x) = \frac{\ln x}{x}$ for $x > 0$.

- (i) Show that the graph of $y = f(x)$ has a stationary point at $x = e$.

2

- (ii) By considering the gradient on either side of $x = e$, or otherwise, show that the stationary point at $x = e$ is a maximum.

1

Question 30 (7 marks)

A truck is making a 1000 kilometre trip at a constant speed of v km/h.

When travelling at v km/h, the truck uses fuel at a rate of $(6 + \frac{v^2}{50})$ litres per hour.

The truck company pays \$2.00 per litre for fuel and pays each of the two drivers \$35 per hour while the truck is travelling.

- (i) Let the total cost of fuel and the driver's pay for the trip be C dollars.

Show that $C = \frac{82000}{v} + 40v$

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- (ii) The truck must take no longer than 12 hours to complete the trip, and speed limits require that $v \leq 110$.

At what speed v should the truck travel to minimise the cost C ?

4

(you may disregard any change-over time for the drivers to swap).

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[illegible]

ENDS



NORTH SYDNEY BOYS HIGH SCHOOL

2023 YEAR 12 HSC ASSESSMENT TASK 3

Mathematics Advanced

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Outcomes – Section I

[illegible]

Section I

1	2	3	4	5	6	7	8	9	10
C	C	A	D	A	D	B	B	D	C

Allow about 15 minutes for this section. Use the multiple-choice answer sheet

- 1 The period of the function $f(x) = 2 \tan(4x - \frac{\pi}{3})$ is

A $\frac{\pi}{2}$

B $\frac{\pi}{3}$

C $\frac{\pi}{4}$ ✓

D π

- 2 For what values of x is the curve $f(x) = x^4 - 2x^3$ concave down?

A $[0, \frac{3}{2}]$

B $(0, \frac{3}{2})$

C $(0, 1)$ ✓

D $[0, 1]$

- 3 Which expression is the derivative of $\cos^2 3x$ when differentiated with respect to x ?

A $-6 \sin 3x \cos 3x$ ✓

B $-2 \sin 3x \cos 3x$

C $2 \sin 3x \cos 3x$

D $6 \sin 3x \cos 3x$

- 4 The amount M of a drug present in the blood after t hours is given by

$$M = 9t^2 - t^3 \text{ for } 0 \leq t \leq 9.$$

When is the amount of drug in the blood increasing most rapidly?

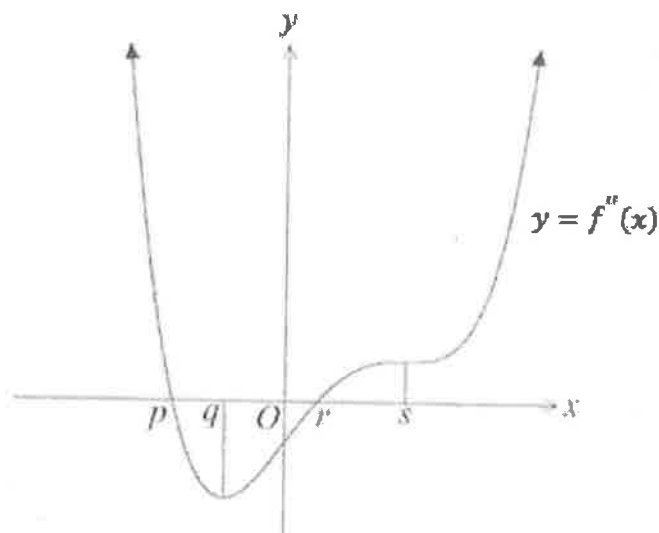
A $t = 0$

B $t = 9$

C $t = 6$

D $t = 3$ ✓

- 5 The diagram shows the graph of $y = f''(x)$ for the function $f(x)$.



For what value of x does the function $f'(x)$ have a maximum turning point?

- A $x = p$ ✓
- B $x = q$
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- D $x = s$

- 6 The table below shows the probability distribution of a discrete random variable X which has mean -0.45 .

x	-2	-1	0	1	2
$P(X = x)$	0.1	a	0.2	0.15	b

What are the values of a and b ?

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- A $(3, 4)$
- B $(-3, 4)$ ✓
- C $(-3, -4)$
- D $(-12, 4)$

- 8 Let X and Y be two events such that $P(X) = 0.5$, $P(Y) = 0.6$, and $P(Y|X) = 0.7$.

Which of the following statements is FALSE?

- A $P(X|Y) < P(Y|X)$
- B X and Y are independent events ✓
- C $P(X \cap Y) = 0.35$
- D $P(X \cup Y) = 0.75$

- 9 What are the domain and range of the function $f(x) = \ln(x+1) - \sqrt{4-x^2}$?

- A domain $(-1, \infty)$ range $(-\infty, \infty)$
- B domain $(-1, \infty)$ range $(-\infty, \ln 3]$
- C domain $(-1, 2]$ range $(-\infty, \infty)$
- D domain $(-1, 2]$ range $(-\infty, \ln 3]$ ✓

- 10 Which of the following is the correct function value at the minimum turning point of $f(x) = (x-2021)(x-2022)(x-2023)$

- A $\frac{1}{\sqrt{3}}$
- B $-\frac{1}{\sqrt{3}}$
- C $-\frac{2\sqrt{3}}{9}$ ✓
- D $\frac{2\sqrt{3}}{9}$

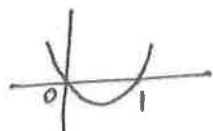
2023 - Y12 Advanced Task 3
Multiple Choice

① $f(x) = 2 \tan(4x - \frac{\pi}{3})$

$\therefore \text{period} = \frac{\pi}{4} \therefore \text{C}$

② $f(x) = x^4 - 2x^3 \therefore f'(x) = 4x^3 - 6x^2$

$\therefore f''(x) = 12x^2 - 12x = 12x(x-1)$



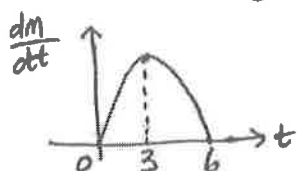
$\therefore 0 < x < 1 \therefore \text{C}$

③ $y = (\cos 3x)^2 \therefore y' = 2 \cos 3x \cdot -3 \sin 3x$

$= -6 \sin 3x \cos 3x \therefore \text{A}$

④ $M = 9t^2 - t^3 \therefore \frac{dM}{dt} = 18t - 3t^2$

$= 3t(6-t)$



$\therefore \text{at } t=3 \therefore \text{D}$

⑤ Graph is $y = f''(x)$. For $f'(x)$ to have max. turning point, we want $f''(x) = 0$ and f'' changing from + to - $\therefore x=p \therefore \text{A}$

⑥ Probs. add to 1, $\therefore a+b = 0.55 \dots (1)$

$E(x) = -0.45 \therefore -0.2 - a + 0 + 0.15 + 2b = -0.45$

i.e. $-a + 2b = -0.40 \dots (2)$

① + ② $\rightarrow 3b = 0.15$

$\therefore b = 0.05$

$a = 0.5$

$\therefore \text{D}$

- ⑦ $y = f(x)$
 Curve has been
 • compressed by factor of 2 $\rightarrow f(2x)$
 • reflected across y axis $\rightarrow f(-2x)$
 • reflected across x axis $\rightarrow -f(-2x)$
 $\therefore (6, -4) \rightarrow (3, -4) \rightarrow (-3, -4) \rightarrow (-3, 4)$ $\therefore$ (B)

- ⑧ $P(Y|X) = \frac{P(Y \cap X)}{P(X)} \therefore 0.7 = \frac{P(Y \cap X)}{0.5}$
 $\therefore P(Y \cap X) = 0.35$
 But $P(Y) \cdot P(X) = (0.6)(0.5) = 0.30 \neq P(Y \cap X)$
 $\therefore$ not independent $\therefore$ (B)

Note: $P(X|Y) = \frac{P(X \cap Y)}{P(Y)} = \frac{0.35}{0.6} \approx 0.58 < P(Y|X) \checkmark$
 $P(X \cap Y) = 0.35 \checkmark$
 $P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$
 $= 0.5 + 0.6 - 0.35 = 0.75 \checkmark$

- ⑨ $f(x) = \ln(x+1) - \sqrt{4-x^2}$
 $D: (-1, \infty)$ $R: (-\infty, \infty)$
 $D: [-2, 2] \rightarrow D: (-1, 2]$ $\therefore$ (D)
 $R: [0, 2] \rightarrow R: (-\infty, \ln 3]$

- ⑩ $f(x) = (x-2021)(x-2022)(x-2023)$ is
 just a horizontal translation of
 $f(x) = (x+1)(x)(x-1)$, & so will have same
 y values at min. turning point. Let $f(x) = x(x^2-1) = x^3-x$
 $f'(x) = 3x^2-1 = 0$ when $x = -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$.
 $f''(x) = 6x > 0$ when $x = \frac{1}{\sqrt{3}}$
 $\therefore y = \left(\frac{1}{\sqrt{3}}+1\right)\left(\frac{1}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}-1\right) = \frac{1}{\sqrt{3}}\left(\frac{1}{3}-1\right)$
 $= \frac{-2}{3\sqrt{3}} = -\frac{2\sqrt{3}}{9}$ $\therefore$ (C)

2023 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

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Student Number:

Mathematics Advanced Section II

90 marks

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24				/5							
25								/6			
26				/4							
27				/2							
28					/4						
29									/5		
30						/7					
Total	/5	/2	/3	/16	/7	/15	/7	/6	/8	/21	/90

Question 11 (3 marks)

The third term of an arithmetic series is 32 and the sixth term is 17.

(i) Find the common difference

1

$$T_3 = a + 2d = 32 \text{ ---- (1)}$$

$$T_6 = a + 5d = 17 \text{ ---- (2)}$$

$$\textcircled{1} - \textcircled{2} \rightarrow -3d = 15 \quad \therefore \textcircled{d = -5} \quad \checkmark$$

(ii) Find the sum of the first ten terms.

2

$$\text{From (i), } a = 32 - 2(-5) \therefore \textcircled{a = 42}$$

$$\begin{aligned} \therefore S_{10} &= \frac{10}{2} (2(42) + (10-1)(-5)) \\ &= 5 (84 - 45) \end{aligned}$$

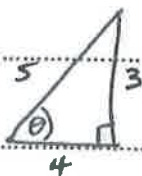
$$\therefore \textcircled{S_{10} = 195} \quad \checkmark \checkmark$$

Question 12 (2 marks)

If $\sin \theta = \frac{3}{5}$ and $\tan \theta < 0$, find the exact value of $\cos \theta$

2

$$\sin > 0, \tan < 0 \therefore \text{Q2} \quad \checkmark \quad \therefore \cos < 0$$



$$\therefore \textcircled{\cos \theta = -\frac{4}{5}} \quad \checkmark \checkmark$$

Question 13 (3 marks)

Given that $2\log_e(x^2y) = 3 + \log_e x - \log_e y$, express y in terms of x in simplest terms. 3

$2\ln(x^2y) = 3 + \ln x - \ln y$ $\therefore 2\ln x^2 + 2\ln y = 3 + \ln x - \ln y$ $4\ln x + 2\ln y = 3 + \ln x - \ln y$ $3\ln x + 3\ln y = 3$ $\ln x + \ln y = 1$ $\ln(xy) = 1$ $\therefore xy = e$ $\therefore y = \frac{e}{x}$	<u>ALT:</u> $\ln x^4 y^2 = 3 + \ln x - \ln y$ $\ln x^4 y^2 - \ln x + \ln y = 3$ $\ln \frac{x^4 y^2 \times y}{x} = 3$ $\ln(x^3 y^3) = 3$ $3\ln xy = 3$ $\ln xy = 1$ $xy = e$ $y = \frac{e}{x}$
--	--

Correctly uses log laws ✓✓

Question 14 (3 marks)

Find the equation of the normal to $y = e^{\cos x}$ at the point where $x = \frac{\pi}{2}$ 3

$$y' = -\sin x \cdot e^{\cos x}$$

When $x = \frac{\pi}{2}$, $y = e^{\cos \frac{\pi}{2}} = e^0 = 1$

$$\& y' = -\sin \frac{\pi}{2} \cdot e^{\cos \frac{\pi}{2}} = -1$$

$$\therefore m_n = 1$$

So normal is $y - 1 = 1(x - \frac{\pi}{2})$

$$\therefore y = x + 1 - \frac{\pi}{2}$$

(ie $x - y + 1 - \frac{\pi}{2} = 0$)

Question 15 (7 marks)

(i) Find $\int \sec^2 3x \, dx$

1

$$= \frac{1}{3} \tan 3x + C \quad \checkmark$$

(Note: maximum penalty 1 mark for lack of constant)

(ii) Find $\int \frac{1}{(3x+2)^4} \, dx$

2

$$\int (3x+2)^{-4} \, dx = \frac{(3x+2)^{-3}}{-3 \times 3} + C$$

$$\therefore = \frac{-1}{9(3x+2)^3} + C \quad \checkmark \checkmark$$

(iii) Evaluate $\int_0^1 e^{-2x} \, dx$ exactly.

2

$$\int_0^1 e^{-2x} \, dx = \left[-\frac{e^{-2x}}{2} \right]_0^1 \quad \checkmark$$

$$= -\frac{e^{-2}}{2} - \left(-\frac{1}{2} \right)$$

$$\therefore = \frac{1}{2} - \frac{1}{2e^2} \quad \checkmark$$

(ie $\frac{e^2 - 1}{2e^2}$)

(iv) Find $\int \frac{x^2}{x^3-5} dx$

2

$$= \frac{1}{3} \int \frac{3x^2}{x^3-5} dx$$

$$= \frac{1}{3} \ln |x^3-5| + C$$

✓ correct answer

✓ absolute value

Question 16 (3 marks)

The curve $y = \sin x$ is stretched horizontally by a factor of 2, then it is shifted $\frac{\pi}{2}$ units right, then it is stretched vertically by a factor of 3 and reflected in the x-axis.

What equation describes the final curve after this sequence of transformations?

3

$$y = \sin x \rightarrow y = \sin \frac{x}{2}$$

✓ horiz-stretch.

$$\rightarrow y = \sin \frac{1}{2} \left(x - \frac{\pi}{2} \right)$$

✓ shift R

$$\rightarrow y = -3 \sin \frac{1}{2} \left(x - \frac{\pi}{2} \right)$$

✓ vert. stretch & reflection

$$\text{(ie } y = -3 \sin \left(\frac{x}{2} - \frac{\pi}{4} \right) \text{)}$$

Question 17 (4 marks)

A new brand of electric bicycle is introduced to the market and 18,000 are sold in the first month. Each month thereafter, the sales are 70% of the sales in the previous month.

- (i) In which month will monthly sales first drop below 1000 per month?

2

We want $T_n = 18000 \times (0.7)^{n-1} < 1000$

$$\therefore 0.7^{n-1} < \frac{1}{18}$$

$$(n-1) \log_{10} 0.7 < \log_{10} \frac{1}{18}$$

$$n-1 > \frac{\log_{10} \frac{1}{18}}{\log_{10} 0.7} \quad \therefore n > 8.10365 + 1$$

$$n > 9.1$$

$\therefore$ in the 10th month.

- (ii) How many bicycles are sold in total in the first year?

1

$$S_{12} = \frac{18000(1 - 0.07^{12})}{1 - 0.07} \approx 59169.52 \dots$$

$\therefore$ 59169 sold in 1st year

- (iii) How many bicycles are eventually sold altogether?

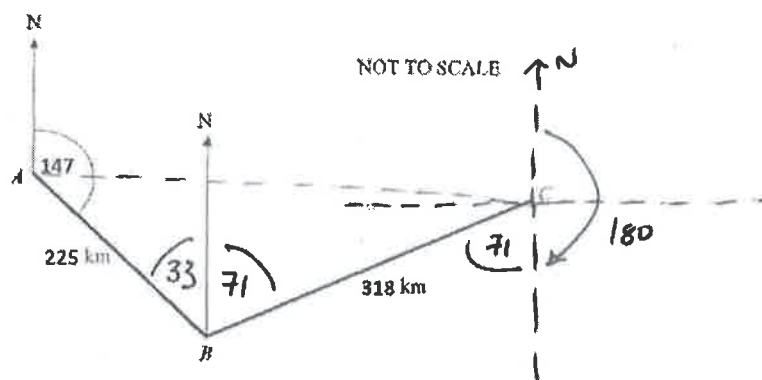
1

as $|r| < 1$ $\therefore$ limiting sum exists

$$\therefore S_{\infty} = \frac{18000}{1 - 0.07} = 60000$$

Question 18 (5 marks)

A ship sails 225 km from Adhiban Island on a bearing of 147 degrees and arrive at Port Bologan to pick up some cattle. It then progresses to its destination, Port Cramling, a distance of 318 km on a bearing of 071 degrees.



- (i) Show that $\angle ABC = 104^\circ$
(you may write on the diagram above)

1

$$\angle ABC = (180^\circ - 147^\circ) + 71^\circ = 33^\circ + 71^\circ = 104^\circ$$

↑
alternate
angle in
parallel
lines

↑
bearing of
C from B

- (ii) Show that the distance AC is approximately 431.7 km.

2

$$AC^2 = 225^2 + 318^2 - 2(225)(318) \cos 104^\circ$$

$$\div 186\,368.0233 \dots$$

$$\therefore AC \div 431.7036 \dots$$

Thus $AC = 431.7 \text{ km (1 d.p.)}$

- (iii) The return trip is a straight line back to Adhiban Island and not passing through Port Bologan. Find the bearing that the ship must take to go straight from Port Cramling to Adhiban Island.

2

$$\frac{\sin \angle ACB}{225} = \frac{\sin 10^\circ}{431.7036}$$

$$\therefore \sin \angle ACB \doteq 0.505709296 \dots$$

$$\therefore \angle ACB \doteq 30^\circ 23'$$

$$\therefore \text{Bearing} = 180^\circ + 71^\circ + 30^\circ 23'$$

$$= \boxed{281^\circ 23' \text{ T.}}$$

Question 19 (5 marks)

Mischa likes to drink pearl milk tea at work. The number X of teas she drinks each day is a random variable with probability distribution given by:

x	0	1	2	3
$P(X=x)$	0.1	0.2	0.3	0.4

- (i) What is the expected value $E(X)$?

1

$$E(X) = 0(0.1) + 1(0.2) + 2(0.3) + 3(0.4)$$

$$= 0 + 0.2 + 0.6 + 1.2 = \boxed{2}$$

- (ii) What are the variance, $\text{Var}(X)$, and the standard deviation, σ ?

2

$$\text{Var}(X) = E(X^2) - \mu^2$$

$$= (0.1)(0^2) + 0.2(1^2) + 0.3(2^2) + 0.4(3^2) - 2^2$$

$$= 5 - 4$$

$$\therefore \boxed{\text{Var}(X) = 1, \sigma = \sqrt{1} = 1}$$

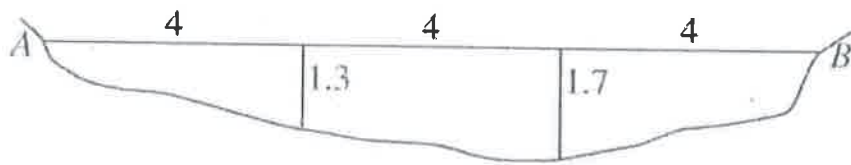
- (iii) Mischa is at work on two successive days. What is the probability that she drinks the same number of pearl milk teas on both days?

2

$$\begin{aligned}
 \text{probability} &= P(0,0) + P(1,1) + P(2,2) + P(3,3) \\
 &= (0.1)^2 + (0.2)^2 + (0.3)^2 + (0.4)^2 \\
 &= 0.01 + 0.04 + 0.09 + 0.16 \\
 \therefore \text{probability} &= 0.3
 \end{aligned}$$

Question 20 (4 marks)

The diagram below shows the cross-section of a stream with the depths of the stream shown in metres at 4 metre intervals. The creek is 12 metres wide.



- (i) Use the trapezoidal rule to approximate the area of the cross-section.

2

x	0	4	8	12
y	0	1.3	1.7	0

$$A \doteq \frac{12-0}{2 \times 3} [0 + 0 + 2(1.3 + 1.7)] = 12 \text{ m}^2$$

- (ii) If water flows through this part of the stream at a speed of 0.5 metres/sec, calculate the approximate volume of water that flows past this section in 1 hour.

2

$$\begin{aligned}
 V &= A \times \text{length} \\
 \text{length} &= \text{speed} \times \text{time} \\
 &= 0.5 \times (60 \times 60) \text{ m} \\
 &= 1800 \text{ m.}
 \end{aligned}$$

$$\begin{aligned}
 \therefore V &\doteq 12 \times 1800 \text{ m}^3 \\
 V &\doteq 21600 \text{ m}^3 \\
 &(\doteq 21600 \text{ KL})
 \end{aligned}$$

Question 21 (8 marks)

Consider the function $y = x^3 - 9x^2 + 24x$.

- (i) Find all stationary points and determine their nature.

4

$$y = x^3 - 9x^2 + 24x$$

$$\therefore y' = 3x^2 - 18x + 24$$

$$= 3(x^2 - 6x + 8)$$

$$y' = 3(x-4)(x-2)$$

For stationary points, $y' = 0 \quad \therefore (x-4)(x-2) = 0$

$$\therefore x = 4$$

$$y = 20$$

or

$$x = 2$$

$$y = 16$$

$$x = 4, x = 2$$

$$y'' = 6x - 18 = 6(x-3)$$

At $(4, 20)$, $y'' = 6(4-3) = 6 > 0$

✓ for testing.

$\therefore$ minimum turning point at $(4, 20)$ ✓
min. point

At $(2, 16)$, $y'' = 6(2-3) = -6 < 0$

$\therefore$ maximum turning point at $(2, 16)$ ✓
max. point.

(ii) Find the point of inflection.

2

For inflections, $y'' = 0 \quad \therefore 6/(x-3) = 0$

$\therefore x = 3, y = 18$ ✓ point

testing. ✓

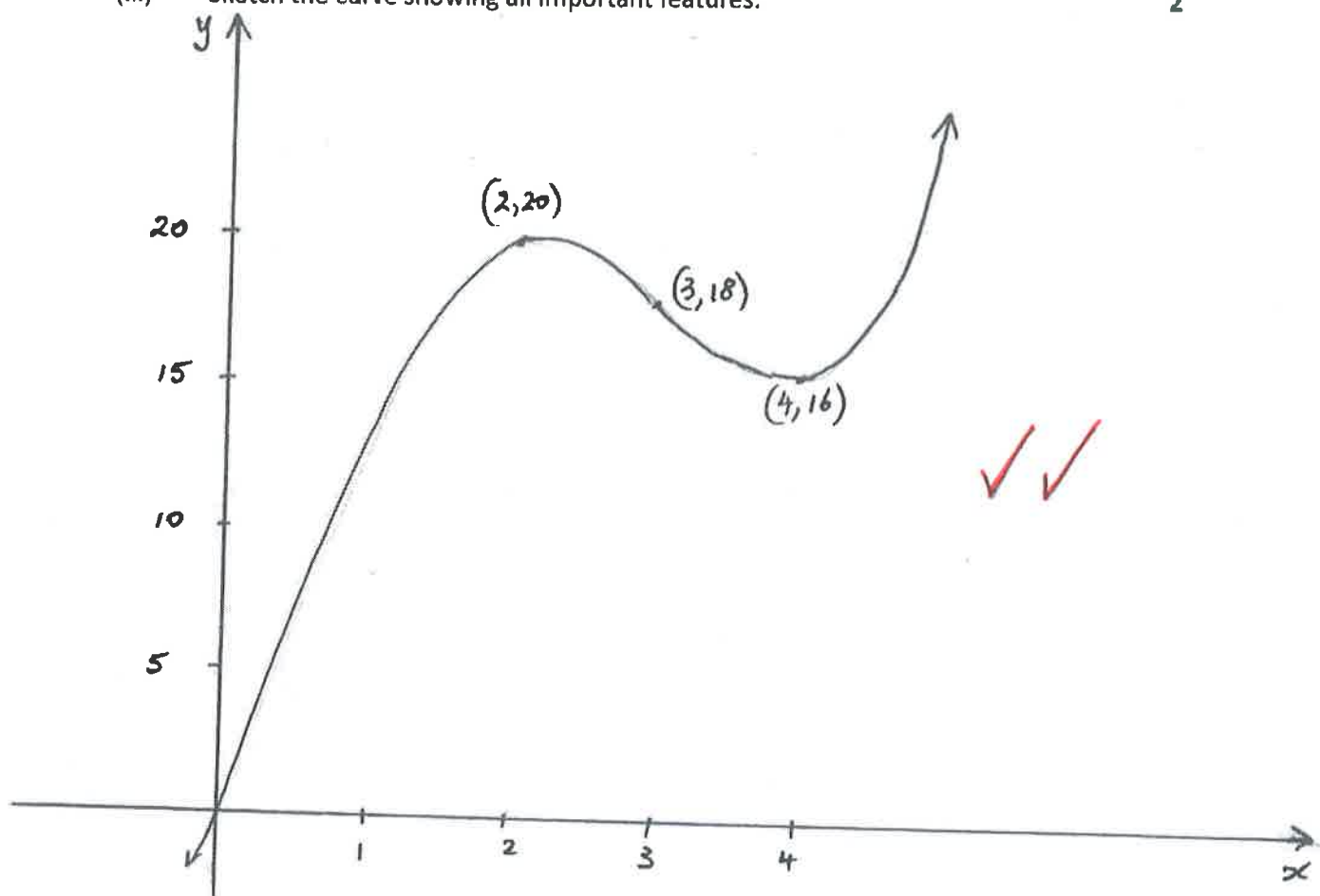
x	2	3	4
y''	-6	0	6

 $\therefore$ change in concavity at $x = 3$

$\therefore (3, 18)$ is point of inflection.

(iii) Sketch the curve showing all important features.

2



Question 22 (5 marks)

The line L is the tangent to the curve $y = x^3 + 7$ at $x = 2$.

- (i) Show that the equation of the tangent L is $y = 12x - 9$

$$y' = 3x^2$$

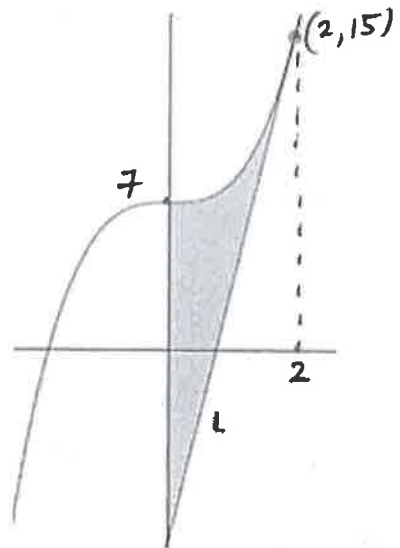
$$\therefore \text{At } x=2, y' = 12 \text{ and } y = 15 \quad \checkmark$$

$\therefore$ tangent equation is

$$y - 15 = 12(x - 2) \quad \checkmark$$

$$y - 15 = 12x - 24$$

$$\therefore y = 12x - 9, \text{ as required.}$$



- (ii) Find the area bounded by the y-axis, the tangent L, and the curve $y = x^3 + 7$

$$A = \int_0^2 [x^3 + 7 - (12x - 9)] dx \quad \checkmark$$

$$= \int_0^2 (x^3 - 12x + 16) dx$$

$$= \left[\frac{x^4}{4} - 6x^2 + 16x \right]_0^2 \quad \checkmark$$

$$= \left[\frac{16}{4} - 6(4) + 16(2) \right] - [0 - 0 + 0]$$

$$\therefore A = 12 \text{ u}^2 \quad \checkmark$$

Question 23 (5 marks)

(i) Show that $\frac{d}{dx}(x \ln x - x) = \ln x$

1

$$\frac{d}{dx}[x \ln x - x] = \ln x \cdot 1 + x \cdot \frac{1}{x} - 1 = \ln x + 1 - 1 = \ln x, \text{ as required.}$$

(ii) Hence or otherwise find $\int \ln x^2 dx$

1

$$\ln x^2 = 2 \ln x$$

$$\therefore \int \ln x^2 dx = 2(x \ln x - x) + C, \text{ from (i).}$$

(iii) The graph shows the curve $y = \ln x^2$, ($x > 0$) which meets the line $x = 5$ at Q. 3

Using your answers from parts (i) and (ii), or otherwise,

find the area of the shaded region.

Area of rectangle OPQR

$$= 5 \times \ln 25$$

$$= 5 \times \ln 5^2$$

$$\therefore \text{Area OPQR} = 10 \ln 5 \text{ u}^2.$$

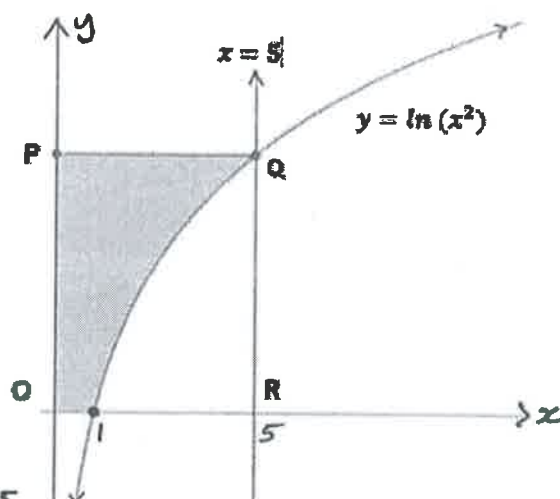
$$\therefore \text{Area shaded region} = 10 \ln 5 - \int_1^5 \ln x^2 dx$$

$$\text{Thus Area} = 10 \ln 5 - 2 \left[x \ln x - x \right]_1^5$$

$$= 10 \ln 5 - 2 \left[(5 \ln 5 - 5) - (\ln 1 - 1) \right]$$

$$= 10 \ln 5 - (10 \ln 5 - 8)$$

$$\therefore \text{Area} = 8 \text{ u}^2$$



Question 24 (5 marks)

A six-sided die is biased so that the number 5 occurs twice as often as any other number.

- (i) The die is rolled once. Show that the probability that an odd number occurs $\frac{4}{7}$.

1

Sample space is $\{1, 2, 3, 4, 5, 5, 6\}$
(these elements are equally likely: we give two 5's since it is twice as likely as the others)

Event is $\{1, 3, 5, 5\}$
 $\therefore P(\text{odd}) = \frac{4}{7}$ ✓

- (ii) If the biased die is rolled twice, find the probability that the sum of the uppermost numbers is seven.

2

$E = \{(2, 5), (5, 2), (3, 4), (4, 3), (1, 6), (6, 1)\}$ ✓
 $\therefore P(E) = \frac{1}{7} \times \frac{2}{7} + \frac{2}{7} \times \frac{1}{7} + 4 \times \left(\frac{1}{7} \times \frac{1}{7}\right)$
 $= \frac{8}{49}$ ✓

The biased die is now rolled together with TWO fair six-sided dice.

(iii) What is the chance that at least two odd numbers are uppermost?

2

"At least 2 odd numbers" means

"3 odd numbers" or "2 odd and 1 even".

Now,

$$P(\text{odd number on fair die}) = \frac{1}{2}, \quad P(\text{odd n}^\circ \text{ on biased die}) = \frac{4}{7}$$

$$P(\text{even number on fair die}) = \frac{1}{2}, \quad P(\text{even n}^\circ \text{ on biased die}) = \frac{3}{7}$$

$$\therefore P(E) = \frac{1}{2} \times \frac{1}{2} \times \frac{4}{7} + P[(O, O, E), \text{ or } (O, E, O) \text{ or } (E, O, O)]$$

biased die biased die

$$= \frac{1}{7} + \frac{4}{7} \times \frac{1}{2} \times \frac{1}{2} + \frac{4}{7} \times \frac{1}{2} \times \frac{1}{2} + \frac{3}{7} \times \frac{1}{2} \times \frac{1}{2}$$

$$= \frac{1}{7} + \frac{11}{28}$$

$$P(E) = \frac{15}{28}$$



Question 25 (6 marks)

The outside temperature (in degrees Celsius) on a certain day was modelled

by $T = 12 + 7\sin\left(\frac{\pi t}{12}\right)$ where t is the number of hours after 6am.

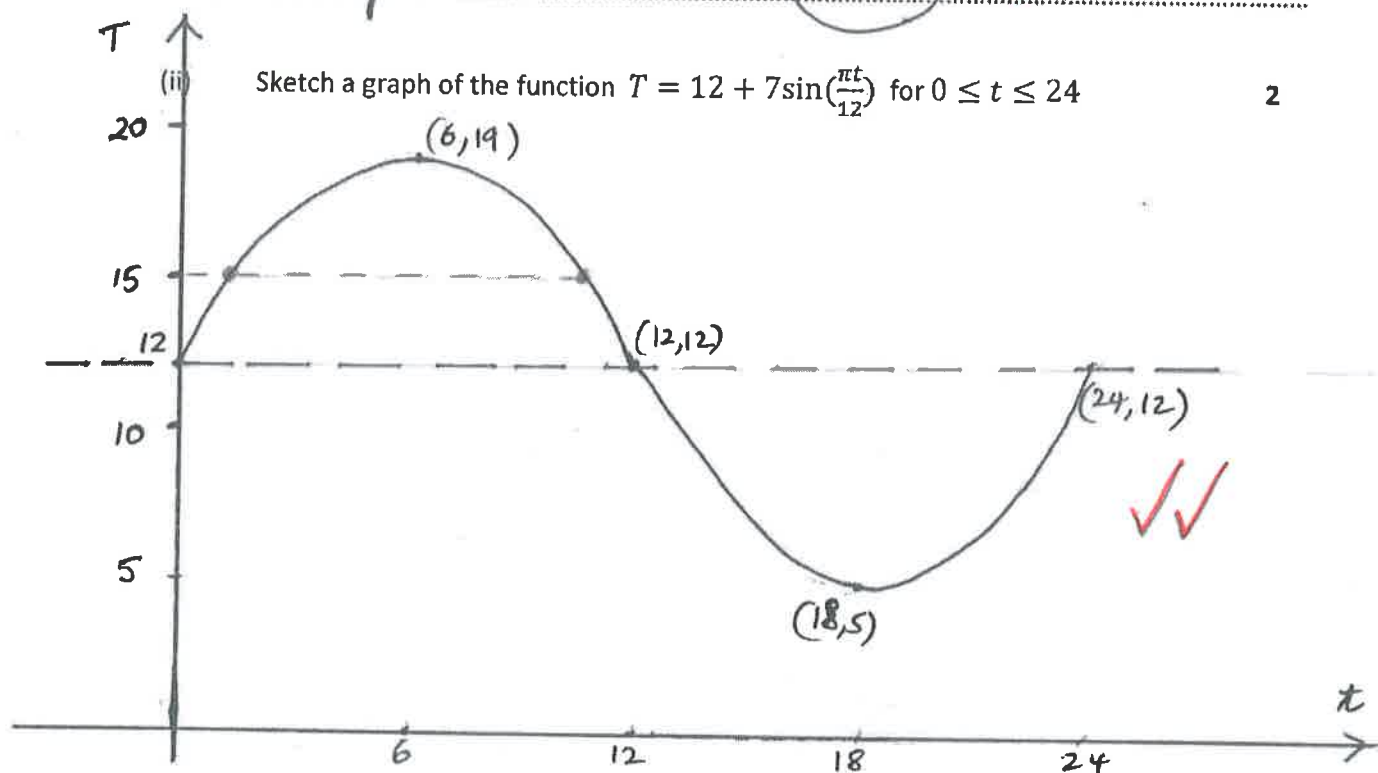
- (i) What is the maximum temperature in the day?

1

Max. temp. = $12 + 7 = 19^\circ\text{C}$ ✓

- (ii) Sketch a graph of the function $T = 12 + 7\sin\left(\frac{\pi t}{12}\right)$ for $0 \leq t \leq 24$

2



- (iii) Between what times during the day is the temperature 15°C or above?

3

Let $12 + 7\sin\left(\frac{\pi t}{12}\right) = 15$

$\therefore \sin\left(\frac{\pi t}{12}\right) = \frac{3}{7}$

Thus $t = \frac{12}{\pi} \sin^{-1} \frac{3}{7}$ or $t = \frac{12}{\pi} \left(\pi - \sin^{-1} \frac{3}{7}\right)$ ✓

$\therefore t \doteq 1.692$ hours or 10.308 hours ✓

ie times are 7.692 or 16.308

$\therefore \geq 15^\circ\text{C}$ from 7:42am and 16:18 pm. ✓

Question 26 (4 marks)

Mrs McCrone walks her three labradoodles at Balmoral Beach every Saturday morning. The dogs are poorly behaved: if a stranger pats them, the chance that the white dog bites him is $\frac{1}{20}$, the chance that the brown one bites him is $\frac{1}{10}$, and the chance that the deranged black one bites him is $\frac{1}{2}$.

- (i) If a stranger selects one of the dogs at random and pats it, what is the chance they will be bitten? 2

$$P(\text{bitten}) = \frac{1}{3} \times \frac{1}{20} + \frac{1}{3} \times \frac{1}{10} + \frac{1}{3} \times \frac{1}{2} \quad \checkmark$$
$$= \frac{13}{60} \quad \checkmark$$

- (ii) Given that a stranger pats one of the dogs and is bitten, what is the probability that it was the black one they patted? 2

$$P(\text{Black} \mid \text{bitten}) = \frac{P(\text{Black and bitten})}{P(\text{bitten})}$$
$$= \frac{\frac{1}{3} \times \frac{1}{2}}{\frac{13}{60}} \quad \checkmark$$
$$= \frac{10}{13} \quad \checkmark$$

Question 27 (2 marks)

Edward plays a game in which he has a probability p of winning, probability q of losing, and probability r of moving to the next round ($p + q + r = 1$).

What is his probability of eventually winning, in terms of p and q ?

2

Let $W =$ 'win round 1', $L =$ 'lose round 1',

$C =$ 'continue after round 1'.

$$\begin{aligned} \text{The } P(\text{eventually wins}) &= P(\text{eventually wins} | W) \cdot P(W) \\ &\quad + P(\text{eventually wins} | L) \cdot P(L) \\ &\quad + P(\text{eventually wins} | C) \cdot P(C) \\ &= 1 \times p + 0 \times q + r \times p(\text{eventually wins}), \end{aligned}$$

since $p(\text{eventually wins} | C) = p(\text{eventually wins})$.

$$\therefore P(\text{eventually wins}) = p + r \times p(\text{eventually wins})$$

$$\therefore p(\text{eventually wins}) = \frac{p}{1-r}$$

$$= \frac{p}{p+q}$$



Question 28 (4 marks)

The point $A(6, 1)$ lies on $h(x)$. The tangent at A is $y = \frac{x}{6}$. Point B is the image of A on the function $g(x) = 3h(2x + 4)$.

(i) Show that B has coordinates $(1, 3)$.

1

$$g(x) = 3h(2x+4) = 3h[2(x+2)]$$

$\therefore$ for any point (x, y) on $h(x)$, the image on

$g(x)$ is $(\frac{x}{2} - 2, 3y)$

$$\therefore B = (\frac{6}{2} - 2, 1 \times 3) = (1, 3)$$



(ii) Hence find the equation of the tangent to $g(x)$ at B.

3

Method 1: tangent to $h(x)$ at A is $y = \frac{x}{6}$

$\therefore$ gradient of tangent to $h(x)$ at A is $\frac{1}{6}$

Thus gradient of tangent to $g(x)$ at B is

$m = \frac{1}{6}$: vertically dilated by factor of 3 &

horizontally dilated by factor of $\frac{1}{2}$

$$\therefore m_B = \frac{1}{6} \times \frac{3}{\frac{1}{2}} = 1 \quad \checkmark \checkmark$$

$\therefore$ tangent to $g(x)$ at B is $y - 3 = 1(x - 1)$

i.e. $y = x + 2$ (or $x - y + 2 = 0$) } $\checkmark$

Method 2:

$$g(x) = 3h(2x+4)$$

$$\therefore g'(x) = 3h'(2x+4) \times 2 = 6h'(2x+4)$$

Since $g'(1) =$ gradient of tangent at B,

$$\therefore g'(1) = 6h'[2(1)+4]$$

$$= 6h'(6)$$

$$= 6 \times \text{gradient of tangent to } h(x) \text{ at A}$$

$$= 6 \times \frac{1}{6}$$

$$= 1$$

$\therefore$ tangent to $g(x)$ at B is $y - 3 = 1(x - 1)$

i.e. $y = x + 2$.

Question 29 (5 marks)

Consider the function $f(x) = \frac{\ln x}{x}$ for $x > 0$.

- (i) Show that the graph of $y = f(x)$ has a stationary point at $x = e$.

2

$$y' = \frac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2} = \frac{1 - \ln x}{x^2}$$

Stationary point means $y' = 0 \therefore 1 - \ln x = 0$

$$\therefore \ln x = 1 \therefore x = e$$

So $\therefore$ stationary point at $x = e$.

- (ii) By considering the gradient on either side of $x = e$, or otherwise, show that the stationary point at $x = e$ is a maximum.

1

x	2.71	e	2.72
y'	0.0004	0	-0.00009

$\therefore$ max. at $x = e$.

$$[\text{alt. : } y'' = \frac{(x^2)(-\frac{1}{x}) - (1 - \ln x)(2x)}{x^4} = \frac{-3x + 2x \ln x}{x^4}]$$

$$\therefore y''(e) = -0.0498 < 0 \therefore \text{max at } x = e$$

(iii) Hence deduce that $e^x \geq x^e$ for all $x > 0$.

2

From (ii), $\frac{\ln x}{x}$ ($x > 0$) has a maximum at $x = e$.

$$f(e) = \frac{\ln e}{e} = \frac{1}{e}$$

$$\text{Thus } \frac{\log_e x}{x} \leq \frac{1}{e}, \quad x > 0$$

$$\therefore \log_e x \leq \frac{x}{e} \quad (\text{as } x > 0)$$

$$e \log_e x \leq x$$

$$\therefore \log_e x^e \leq x$$

$$\therefore e^{\log_e x^e} \leq e^x$$

$$\therefore x^e \leq e^x \quad (\text{as } e^{\log_e a} = a)$$

$$\text{i.e. } e^x \geq x^e$$



Question 30 (7 marks)

A truck is making a 1000 kilometre trip at a constant speed of v km/h.

When travelling at v km/h, the truck uses fuel at a rate of $(6 + \frac{v^2}{50})$ litres per hour.

The truck company pays \$2.00 per litre for fuel and pays each of the two drivers \$35 per hour while the truck is travelling.

- (i) Let the total cost of fuel and the driver's pay for the trip be C dollars.

Show that $C = \frac{82000}{v} + 40v$

3

$$\text{Time for trip} = \frac{1000}{v} \text{ hours}$$

$$\therefore \text{Driver cost} = 2 \times \$35 \times \frac{1000}{v} = \$ \frac{70000}{v}$$

$$\begin{aligned} \text{Fuel cost} &= \$2 \times \left(6 + \frac{v^2}{50}\right) \times \frac{1000}{v} = \frac{2000}{v} \left(6 + \frac{v^2}{50}\right) \\ &= \frac{12000}{v} + 40v \end{aligned}$$

$$\therefore C = \frac{70000}{v} + \frac{12000}{v} + 40v$$

$$\therefore C = \frac{82000}{v} + 40v \text{ dollars.}$$

- (ii) The truck must take no longer than 12 hours to complete the trip, and speed limits require that $v \leq 110$.

At what speed v should the truck travel to minimise the cost C ?

4

(you may disregard any change-over time for the drivers to swap).

$$C = 82000 v^{-1} + 40v$$

$$\therefore \frac{dC}{dv} = -\frac{82000}{v^2} + 40$$

$$\text{For min, } \frac{dC}{dv} = 0 \therefore \frac{82000}{v^2} = 40 \therefore v^2 = 2050$$

$$\therefore v = \sqrt{2050} \approx 45.3 \text{ km/h.}$$

But, if $v = \sqrt{2050}$, $t = \frac{1000}{\sqrt{2050}} \div 22.1$ hours

This is too long, as we need $t \leq 12$ (ie. $v \geq 83\frac{1}{3}$)

In fact, the domain here is $83\frac{1}{3} \leq v \leq 110$.

So we need to check the cost at the endpoints of the domain.

$$\text{If } v = 83\frac{1}{3}, C = \frac{82000}{83\frac{1}{3}} + 40(83\frac{1}{3}) \div \$4317$$

$$\text{If } v = 110, C = \frac{82000}{110} + 40(110) \div \$5145$$

So we get minimum cost when $v = 83\frac{1}{3}$ km/h.

[Since $\frac{d^2C}{dv^2} = 2 \times 82000 v^{-3} > 0$ for all $v > 0$

So $v = \sqrt{2050}$ gives min. turning point &

C is increasing after $v = \sqrt{2050}$. As v increases,

C increases, so the minimum C that satisfies the conditions corresponds to $v = 83\frac{1}{3}$ km/h.

But all that's required is to calculate C at the endpoints.]

ENDS